



MEDICAL IMAGING—RADIATION ONCOLOGY—REVIEW ARTICLE OPEN ACCESS

AI Revolution in Radiology, Radiation Oncology and Nuclear Medicine: Transforming and Innovating the Radiological Sciences

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ABSTRACT

The integration of artificial intelligence (AI) into clinical practice, particularly within radiology, nuclear medicine and radiation oncology, is transforming diagnostic and therapeutic processes. AI-driven tools, especially in deep learning and machine learning, have shown remarkable potential in enhancing image recognition, analysis and decision-making. This technological advancement allows for the automation of routine tasks, improved diagnostic accuracy, and the reduction of human error, leading to more efficient workflows. Moreover, the successful implementation of AI in healthcare requires comprehensive education and training for young clinicians, with a pressing need to incorporate AI into residency programmes, ensuring that future specialists are equipped with traditional skills and a deep understanding of AI technologies and their clinical applications. This includes knowledge of software, data analysis, imaging informatics and ethical considerations surrounding AI use in medicine. By fostering interdisciplinary integration and emphasising AI education, healthcare professionals can fully harness AI's potential to improve patient outcomes and advance the field of medical imaging and therapy. This review aims to evaluate how AI influences radiology, nuclear medicine and radiation oncology, while highlighting the necessity for specialised AI training in medical education to ensure its successful clinical integration.

Abbreviations: AI, artificial intelligence; DL, deep learning; ML, machine learning; NM, nuclear medicine; RO, radiation oncology; RT, radiation therapy.

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1 | Introduction

Artificial intelligence (AI) integration in the radiological area has emerged as a transformative force, promising to reshape the landscape of diagnostic imaging and radiation oncology (RO). Recent advancements in AI, particularly in deep learning (DL) and machine learning (ML) algorithms, have significantly enhanced image recognition, interpretation and decision-making processes [1–4]. This is a rapid transformation from a discipline based on the visual interpretation of the images toward a new type of imaging, which integrates the quantitative data (biomarkers) coming from the images with the interpretative modality, since the extracted quantitative features can be analysed to reflect the underlying pathophysiology [5, 6]. As a result, AI is increasingly being recognised not just as a complementary tool but as a critical component in the clinical workflow [7].

The impetus for this review stems from a recent meeting at the first joint congress, the so-called ‘Next Generation Congress’, of the Italian radiological area held in Milan in June 2024, which brought together young residents and specialists in radiology, radiation oncology and nuclear medicine (NM). This gathering underscored the growing relevance of AI in clinical practice and highlighted the urgent need for its widespread dissemination and adoption, particularly among the next generation of practitioners [8–10].

The discussions emphasised that, while AI has already begun to make significant inroads into the field, its full potential remains largely untapped, particularly in routine clinical settings. In current diagnostic and therapeutic clinical practice, AI’s most notable contributions include the automation of image analysis, enhancement of diagnostic accuracy and reduction of human error [11]. For instance, AI algorithms have demonstrated superior performance in detecting pathologies such as pulmonary nodules, breast lesions and intracranial haemorrhages, often with speed and precision surpassing human radiologists [12]. Furthermore, AI has shown potential to improve workflow efficiency by automating routine tasks such as image sorting, triage, and even preliminary reporting of some diagnostic examinations, allowing radiologists to focus on more complex cases [13–15].

Despite these advances, the adoption of AI in daily clinical practice is challenging. There is an ongoing debate about the ethical implications of AI, particularly concerning data privacy, the potential for algorithmic bias, and the overarching question of whether AI might one day supplant the physician [16]. However, the consensus among experts is that AI is more likely to augment rather than replace human expertise, serving as a powerful adjunct in the diagnostic process [17–19].

A critical aspect of this transition is educating and training young residents and specialists in the radiological area. The next generation must be equipped not only with traditional skills but also with a deep understanding of AI technologies and their applications. This requires an updated curriculum integrating AI into radiology, RO and NM training programmes, fostering a generation of specialists adept at using these tools to enhance patient care [20–22].

This review aims to provide an overview of the current state of AI in radiology, RO and NM, highlight its practical applications, and discuss strategies for fostering AI adoption among specialists.

2 | The Importance of Teaching AI During Residency

Before exploring the specific reasons why AI should be taught to radiology, NM and RO residents, it is essential to understand the key advantages of AI in medical education. The development of Generative Artificial Intelligence (GenAI) models, such as Large Language Models (LLMs), is creating a disruption in educational landscapes. GenAI offers significant opportunities to personalise educational programmes, automate routine tasks, create tailored content, and enhance access to evidence-based information that could fill the gaps in resident clinical preparation, hypothetically leading to a reduction in the time needed to prepare the trainees adequately [23, 24]. However, the integration of GenAI into medical education raises critical challenges and concerns about data privacy, accuracy, and the risk of over-reliance on AI, highlighting the importance of equipping the younger generation of clinicians, particularly radiologists, nuclear medicine physicians and radiation oncologists, with a robust understanding of AI’s strengths and limitations [24]. Addressing these issues and preparing future healthcare professionals to navigate AI’s potential and risks will be essential for the responsible and effective implementation of AI in healthcare.

3 | The Importance of Teaching AI to Young Radiologists

The rapid spread of AI in clinical practice creates an urgent need to improve the education and training of young radiologists to fully understand the potential and advantages of AI applications in patient clinical care [25–27]. Implementing AI by young radiologists requires basic knowledge of software use, data analysis and imaging informatics fundamentals, data safety, potential biases, ethical considerations and evolving regulations of AI [22, 28, 29]. In addition, the necessity of tailored educational programmes is an emerging demand among young radiologists [30, 31]. Most young radiologists and trainees believe that AI basics should be taught during residency, but the actual implementation in residency programmes remains scant [32, 33]. In a multi-institutional survey, 83% of residents in the United States agree that AI education should be part of the radiology residency curriculum [34]. Similarly, in an international survey, 79% of radiologists and radiology residents reported that AI should be incorporated into residency programmes, and lack of knowledge was considered a hurdle for AI implementation by 57% of the responders [35]. In a survey among members of the SIRM Lombardy, young radiologists under the age of 30 considered themselves not adequately informed on the AI topics, with significant difference of opinion compared to older radiologists [36].

Most of the current education on AI in radiology is based on conference meetings, scientific literature, workshops by commercial

companies, and dedicated online training. Particularly, commercial companies worldwide offer a large variety of AI theoretical or hands-on courses to improve young radiologists' knowledge and skills, while the involvement of academic institutions is still limited [37, 38]. More structured training should be provided during the radiology residency to ensure an impartial and industry-unbiased education of young radiologists [39–41]. In this context, the training curriculum of the European Society of Radiology recommends teaching basic principles of medical imaging informatics and AI tools during general radiology training (level I–II) and learning knowledge of clinical applications of AI during subspecialization in radiology (level III) [42].

4 | The Importance of Teaching AI to Young Nuclear Medicine Physicians

Much like radiology, NM has experienced a rapid evolution with the integration of AI into clinical practice. This advancement encompasses improvements in instrumentation, image acquisition and formation, image reconstruction, low-dose and fast scanning techniques, imaging quantification, image interpretation—including computer-aided detection, diagnosis and prognosis—as well as internal radiation dosimetry for radioligand therapy [43–49].

The convergence of AI and NM has created a synergistic relationship that promises groundbreaking opportunities for the future. However, to fully realise this potential, overcome current limitations and keep pace with the swift progression of AI technologies, it is essential to foster a controlled cultural shift toward innovation through comprehensive training and education [10, 50]. Disseminating knowledge about implementing AI in NM and molecular imaging is greatly needed, and educating young physicians in this area will be crucial for the sustained advancement of the discipline.

Effective implementation of AI must be accompanied by robust educational initiatives to ensure that no one is left behind. This requires a thorough understanding of software, data analysis, imaging informatics, data safety, potential pitfalls, ethical considerations, and the evolving regulations surrounding AI [50, 51]. In this context, a significant effort should be made by academia to adapt the training programme to this evolving scenario by organising dedicated courses involving experts in the field who can adequately teach and train students and physicians in this new avenue of NM. Developing educational programmes within NM training schools that address scientific and clinical aspects will help current and future scientists and clinicians become familiar with and adept at using AI, enhancing its adoption and development.

The European School of Multimodality Imaging and Therapy (ESMIT), the educational branch of the European Association of Nuclear Medicine (EANM), has already initiated efforts in this direction by offering webinars and advanced courses at the EANM Educational Facility in Vienna, such as the 2023 course titled 'Artificial Intelligence in Imaging: Making the Future' [45]. This starting commitment to education will ensure that NM remains at the forefront of medical innovation, well-prepared to meet the challenges and seize the opportunities presented by AI advancements [52].

5 | The Importance of Teaching AI to Young Radiation Oncologists

AI is revolutionising many fields of medicine, including RO, where precision and efficiency are critical aspects [53–56]. AI can assist in processing medical images and planning treatments, allowing for more accurate identification and localisation of tumours. This helps personalise therapies, minimise damage to healthy tissues, and increase the treatment's effectiveness [57–59].

The impact of AI on radiation therapy (RT) is significant, with the potential for AI and ML to excel in various aspects of the RT process. These technologies not only improve the quality of training for residents but are also becoming an increasingly essential resource in the medical field [60, 61].

A recent systematic review by the Young Group of the Italian Association of Radiotherapy and Clinical Oncology (yAIRO) highlighted how AI and radiomics seem to significantly impact image-guided radiation therapy (IGRT) in all the phases of RT workflow [62]. Young radiation oncologists expressed both enthusiasm and concern about automatic segmentation, as revealed by a survey conducted by the SFjRO (Société Française des jeunes Radiothérapeutes Oncologues) [63]. Many participants supported incorporating these tools into training to ensure future radio-oncologists maintain a strong knowledge of radio-anatomy. However, one-third of respondents suggested restricting AI tools to senior physicians to prevent neglect of this essential knowledge [63].

Guckenberger et al. evaluated the potential of the AI chatbot ChatGPT in supporting young clinical scientists. ChatGPT was generally favoured by less experienced researchers, but it did not bridge the gap in scientific expertise; in fact, less experienced researchers were more likely to accept the false results produced uncritically by the AI [64].

Training young radiation oncologists on AI is a technical necessity and an essential step to ensure high-quality, safe and leading-edge medicine. Teaching AI to young radiation oncologists is crucial for several reasons, including improving the quality of care, process optimisation, innovation and research, and preparing for the future [60, 65–67].

6 | AI: Clinical Application

6.1 | AI in Radiology: Clinical Application

AI is playing an increasingly transformative role in radiology. To date, a total of 213 AI-based software applications have received CE approval, which is a European certification indicating conformity with health, safety and performance standards; hundreds more are currently under development and validation. The latest comprehensive list of these approved AI tools is available in the official registry, accessible at this link: <https://radiology.healthregister.com> [68–70].

AI algorithms primarily target neuroradiology and chest radiology, followed by breast, abdominal and musculoskeletal

imaging applications [71–78]. AI applications span across all radiological modalities, including radiography, computed tomography (CT), magnetic resonance (MR) and ultrasound (US) [79–83], and can be classified into two categories: interpretative and non-interpretative uses [68, 84]. Broadly speaking, interpretative uses encompass the processes of detection, characterisation and monitoring. Of these, detection applications were the first to be developed, and they remain the most widely validated and implemented AI tools in clinical practice [85].

When examining the areas where AI has advanced the most, neuroradiology stands out, with key applications centred around optimising stroke workflows, detecting large vessel occlusions, quantifying stroke scoring metrics, segmenting and assessing ischaemic or at-risk tissue, and predicting clinical outcomes in stroke patients [86]. In chest radiology, the most prominent applications are in the field of lung cancer, focusing on lesion detection, segmentation, and characterisation [72, 87], as well as in the emergency setting (Figure 1) [88, 89]. In breast radiology, the first area where AI was extensively studied and has achieved the widest global adoption is mammographic screening, which involves lesion detection [90, 91]. Building on the success in these domains, AI applications in breast are now being tested across all imaging modalities, with an increasingly prominent role in lesion characterisation [90, 92, 93]. A list of

AI applications across the main fields of radiology is provided in Table 1.

Non-interpretative uses of AI include tools designed to enhance workflow efficiency, support clinical protocols, optimise scanner and patient scheduling, and facilitate structured radiological reporting through language processing [84, 95, 96]. AI is becoming increasingly integrated into both clinical practice and research. As asserted by Langlotz [97], while it is unlikely that radiologists will be replaced by AI, those who do not adopt AI will undoubtedly be outpaced by those who do. This principle underpins scientific progress across all fields, particularly in healthcare. It also highlights the importance of understanding the foundational principles and ethical considerations necessary for the full and responsible implementation of AI in radiology [98–100].

6.2 | AI in Nuclear Medicine: Clinical Application

As mentioned above, the development of AI represents a breakthrough in NM, providing many applications. Besides the widely spread AI-driven managing pipelines, including patient scheduling, report generation and images management, the main clinical applications of AI in NM involve images generation, post-processing and analysis [45, 101, 102].

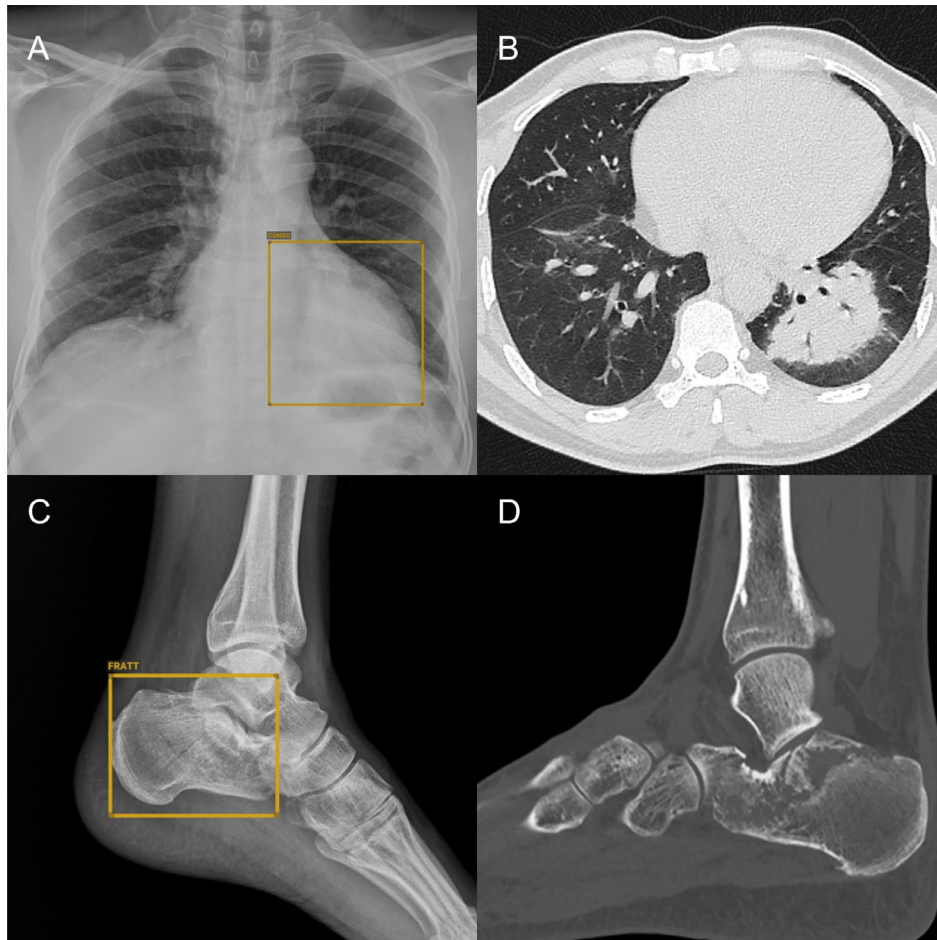


FIGURE 1 | Examples of AI-assisted diagnosis in radiology. Chest radiograph (A) showing a subtle retrocardiac lung consolidation identified by AI (ChestView, Gleamer), and confirmed on CT (B). Bone radiograph (C) acquired in a trauma patient with calcaneus fracture detected by AI (BoneView, Gleamer) and confirmed on CT (D).

Recently, DL and ML models were trained and implemented both in the traditional iterative reconstruction process and post-processing analysis, optimising image quality with significant dose reduction, shortening acquisition time, and extracting imaging biomarkers that may evaluate clinical endpoints. With the spreading of hybrid imaging (PET/CT, PET/MR and SPECT/CT), many studies aimed to develop AI-assisted tools for image attenuation and motion correction, with promising results [43, 46, 103, 104].

Currently, the main improvement gained by AI analysis involves image segmentation, especially in oncologic, cardiac and neurologic NM imaging [103, 104]. In oncologic imaging, the feasibility of AI-assisted image analysis for detecting malignant lesions, automating segmentation, and extracting radiomic features—which may predict disease progression and therapy response—can also estimate whole-body dosimetry in patients referred for radioligand therapy [47, 104] (Figure 2).

The growing use of target-specific PET tracers (e.g., ^{18}F -PSMA-1007 and ^{68}Ga -PSMA in prostate cancer, ^{18}F -FES in breast cancer) allows the development of further algorithms predicting histologic diagnosis and molecular tumour classification [105–109].

Myocardial perfusion imaging (MPI) by SPECT or PET is another broad field of AI application [110]. AI allows nuclear cardiology studies to be acquired with lower doses or shorter acquisition times, also providing synthetic attenuation correction of SPECT images to improve diagnostic accuracy [111, 112]. ML and AI-assisted algorithms also showed a good performance in myocardial segmentation compared to experienced NM physicians. Moreover, ML-extracted quantitative features have good prognostic value in detecting obstructive coronary disease and predicting cardiac adverse events when integrated with clinical data [112–114].

TABLE 1 | A list of AI applications across the main fields of radiology.

Subspecialty	Main applications [94]
Neuroradiology	Detection of strokes, haemorrhages and aneurysms, alongside the assessment and segmentation of brain structures
Chest radiology	Detection of lung nodules, pneumonia, pneumothorax and rib fractures. It also includes the analysis of lung diseases and emphysema quantification
Breast radiology	Lesion detection and characterisation, breast density analysis
Abdominal radiology	Liver segmentation, genitourinary structure analysis, lesion identification, characterisation of focal liver lesions and adrenal lesion characterisation
Musculoskeletal radiology	Detection of fractures and osteoarthritis, and the quantitative imaging of bone structures
Cardiovascular radiology	Coronary artery assessments, myocardial infarction diagnostics and evaluation of cardiomyopathies alongside general cardiac function
Emergency radiology	Rapid identification of haemorrhages, occlusions, fractures and various abdominal emergencies

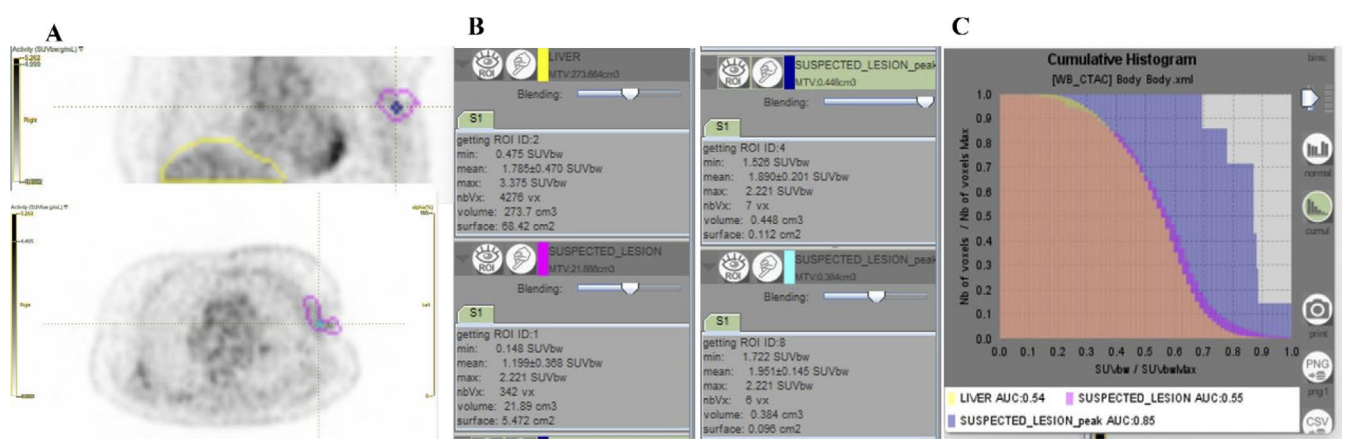


FIGURE 2 | AI-assisted interpretation of an ^{18}F -FDG PET-CT scan. Standard PET images (A) highlight a suspicious area of uptake in the left axilla (pink), with the maximum uptake peak marked (blue circle) and liver uptake shown for reference (yellow). The AI algorithm automatically segmented the regions and quantified uptake values in both the suspicious area and the liver, ultimately suggesting a benign nature (B). Cumulative histograms illustrate the voxel-wise activity distribution across all analysed regions (C). In the specific case, the lesion was initially equivocal, with differential diagnosis including metastatic recurrence of infiltrating ductal carcinoma (pT2N1) versus post-radiotherapy inflammatory changes. Histological examination confirmed granulomatous tissue, consistent with the benign assessment provided by the AI.

Technical innovation in AI has led to considerable advances in brain imaging as well. Along with the already proven usefulness in image quality assessment and segmentation, many authors demonstrated the added value of DL methods for image analysis in identifying neurodegenerative disorders and predicting disease evolution (e.g., ¹²³I-Ioflupane in Parkinson's disease, ¹⁸F-FDG and ¹⁸F-Florbetaben in Alzheimer's disease) [115–120].

AI applications may, therefore, help enhance the role of NM in several clinical scenarios, with promising results. AI enables patient-tailored management in various oncologic, cardiologic and neurological settings. However, further and wider studies are needed to standardise the different algorithms and obtain reliable parameters from different DL models.

6.3 | AI in Radiation Oncology: Clinical Application

RO is closely tied to technological advancements, and the emergence of AI has the potential to revolutionise this field. AI can accelerate processes, improve outcomes, and reduce the workload in daily clinical practice, as shown in Table 2 [54, 121–123]. Xiao et al. used a DL-based tool to enhance the segmentation of organs at risk in cervical cancer [124]. In a study of 313 patients, automatic contouring was compared to manual segmentation by expert radiation oncologists. The AI achieved a Dice similarity coefficient between 0.91 and 0.98 for most organs, with an average segmentation time of 6.6s. The rectum had the poorest performance, likely due to its small size and unclear boundaries [124]. Similarly, Liao et al. developed a DL tool for segmenting 16 organs at risk in the abdomen using 544 CT scans. The tool provided accurate contouring for the liver, kidneys and spleen, with minor errors in organs like the stomach, oesophagus, adrenal glands and rectum. Major revisions were needed only for the colon and small bowel contours [125].

In the field of non-small-cell lung cancer, Hosny et al. conducted an observational study to validate DL models for target segmentation. While the model performed well in internal validation, it did not surpass the benchmark when tested with external data from different experts. However, AI assistance led to a significant 65% reduction in segmentation time and a 32% reduction in inter-observer variability [126]. Zevenino et al. reported the

successful clinical implementation of an auto-planning tool for left-sided breast cancer [127]. The AI system planned 17 clinical treatments, which were then compared to manual plans. Compared to manual planning, the AI achieved less than 1% error and improved the dose distribution to organs at risk, such as the lungs and the left anterior descending artery [127]. During RT, Lee et al. developed a DL algorithm to predict weekly anatomical changes in lung tumours and the oesophagus. This tool accurately predicted structural changes, resulting in a 4.2Gy reduction in the mean oesophageal dose while maintaining adequate tumour coverage [128].

Finally, AI is also proving valuable as a predictive tool. Li et al. created a DL approach to predict clinical outcomes based on magnetic resonance imaging. Applied to 99 rectal cancer patients, this AI model predicted a complete clinical response with an accuracy of 0.789, a specificity of 0.725, and a sensitivity of 0.812 [129].

7 | AI: Unresolved Issues and Emerging Challenges

As AI progresses, its use in healthcare, especially in all radiological fields, requires careful ethical and legal evaluation. To accomplish this goal, the European Union published the so-called ‘AI Act’ in 2024. The act classifies AI applications into four risk levels, with medical use designated as High-Risk [130]. This category includes AI systems that may pose significant threats to health, safety, or fundamental rights, necessitating strict adherence to standards of quality, transparency, human oversight and safety [130].

Another essential aspect that must be considered to properly understand AI functionality is knowing that AI in healthcare presents specific risks, including algorithmic bias, errors and issues, such as AI hallucinations, where models generate incorrect but plausible results [131]. Furthermore, an article by Gichoya et al. explains that bias in medical imaging AI can emerge throughout the entire AI lifecycle, from design to deployment, underscoring the need to create and use diverse and representative datasets, develop and test rigorous validation protocols, and perform ongoing monitoring and evaluation of model performance to minimise these biases [132]. These measures are essential to promote fair and equitable healthcare outcomes through responsible AI development and implementation. Mitigating these risks requires technical solutions and a collaborative approach, engaging clinicians, AI developers, and ethicists to build trustworthy AI systems that enhance rather than undermine the quality of healthcare delivery.

8 | Conclusions and Perspectives

The integration of AI into clinical practice presents a tremendous opportunity to enhance diagnostic accuracy and operational efficiency, particularly in disciplines such as radiology, NM and RO. The success of this transformation depends on the collaboration and integration of these fields, as AI enables more comprehensive and precise approaches to patient care. By combining the strengths of each specialty, AI facilitates the development of Precision Medicine, which should no longer be

TABLE 2 | How artificial intelligence improves radiation oncology.

Function	AI contribution
Contouring	AI provides accurate contouring in a very small time
Planning	AI improves dose distribution to OARs while maintaining dose coverage of the target
Adapting	AI quantifies structural changes, achieving good adaptation to anatomic changes
Predicting	AI predicts clinical responses, leading to more tailored treatment

an unattainable utopia but a realistic and achievable goal [133]. This integration allows for tailored diagnostic and therapeutic strategies, improving outcomes by addressing each patient's unique needs.

A critical component of ensuring this evolution is education. Future physicians must be trained not only in traditional clinical skills but also in the use and understanding of AI technologies. Teaching AI during the training is no longer optional but necessary to equip them to navigate an ever-evolving field. Only through updated and continuous educational programmes can the widespread and safe adoption of AI technologies be guaranteed, enabling next generation professionals to improve patient care and contribute to the advancement of modern medicine.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

References

1. E. Montagnon, M. Cerny, A. Cadrin-Chênevert, et al., "Deep Learning Workflow in Radiology: A Primer," *Insights into Imaging* 11, no. 1 (2020): 22.
2. ESR, "What the Radiologist Should Know About Artificial Intelligence – An ESR White Paper," *Insights into Imaging* 10, no. 1 (2019): 44.
3. L. Pinto-Coelho, "How Artificial Intelligence Is Shaping Medical Imaging Technology: A Survey of Innovations and Applications," *Bioengineering (Basel)* 10, no. 12 (2023): 1435.
4. M. Torres-Velazquez, W. J. Chen, X. Li, and A. B. McMillan, "Application and Construction of Deep Learning Networks in Medical Imaging," *IEEE Transactions on Radiation and Plasma Medical Sciences* 5, no. 2 (2021): 137–159.
5. S. Majumder, S. Katz, D. Kontos, and L. Roshkovan, "State of the Art: Radiomics and Radiomics-Related Artificial Intelligence on the Road to Clinical Translation," *BJR Open* 6, no. 1 (2023): tzad004.
6. R. Najjar, "Redefining Radiology: A Review of Artificial Intelligence Integration in Medical Imaging," *Diagnostics (Basel, Switzerland)* 13, no. 17 (2023): 2760.
7. B. J. Erickson, P. Korfiatis, Z. Akkus, and T. L. Kline, "Machine Learning for Medical Imaging," *Radiographics* 37, no. 2 (2017): 505–515.
8. S. M. Santomartino and P. H. Yi, "Systematic Review of Radiologist and Medical Student Attitudes on the Role and Impact of AI in Radiology," *Academic Radiology* 29, no. 11 (2022): 1748–1756.
9. Y. S. Kwon, M. Dohopolski, H. Morgan, et al., "Artificial Intelligence–Empowered Radiation Oncology Residency Education," *Practical Radiation Oncology* 13, no. 1 (2023): 8–10.
10. A. W. J. M. Glaudemans, R. A. J. O. Dierckx, B. Scheerder, et al., "The First International Network Symposium on Artificial Intelligence and Informatics in Nuclear Medicine: 'The Bright Future of Nuclear Medicine Is Illuminated by Artificial Intelligence'," *European Journal of Nuclear Medicine and Molecular Imaging* 51, no. 2 (2024): 336–339.
11. A. Lenskjold, J. U. Nybing, C. Trampedach, et al., "Should Artificial Intelligence Have Lower Acceptable Error Rates Than Humans?," *BJR Open* 5, no. 1 (2023): 20220053.
12. T. D'Angelo, G. M. Bucolo, T. Kamareddine, et al., "Accuracy and Time Efficiency of a Novel Deep Learning Algorithm for Intracranial Hemorrhage Detection in CT Scans," *Radiologia Medica* 129, no. 10 (2024): 1499.
13. M. P. McBee, O. A. Awan, A. T. Colucci, et al., "Deep Learning in Radiology," *Academic Radiology* 25, no. 11 (2018): 1472–1480.
14. C. A. Mallio, A. C. Sertorio, C. Bernetti, and B. Beomonte Zobel, "Radiology, Structured Reporting and Large Language Models: Who Is Running Faster?," *Radiologia Medica* 128, no. 11 (2023): 1443–1444.
15. Z. Babar, T. van Laarhoven, F. M. Zanzotto, and E. Marchiori, "Evaluating Diagnostic Content of AI-Generated Radiology Reports of Chest X-Rays," *Artificial Intelligence in Medicine* 116 (2021): 102075.
16. F. M. Aldhafeeri, "Navigating the Ethical Landscape of Artificial Intelligence in Radiography: A Cross-Sectional Study of Radiographers' Perspectives," *BMC Medical Ethics* 25, no. 1 (2024): 52.
17. F. Pesapane, M. Codari, and F. Sardanelli, "Artificial Intelligence in Medical Imaging: Threat or Opportunity? Radiologists Again at the Forefront of Innovation in Medicine," *European Radiology Experimental* 2, no. 1 (2018): 35.
18. P. Lakhani, A. B. Prater, R. K. Hutson, et al., "Machine Learning in Radiology: Applications Beyond Image Interpretation," *Journal of the American College of Radiology* 15, no. 2 (2018): 350–359.
19. A. Verghese, N. H. Shah, and R. A. Harrington, "What This Computer Needs Is a Physician," *Journal of the American Medical Association* 319, no. 1 (2018): 19.
20. S. Wang and R. M. Summers, "Machine Learning and Radiology," *Medical Image Analysis* 16, no. 5 (2012): 933–951.
21. E. Neri, G. Aghakhanyan, M. Zerunian, et al., "Explainable AI in Radiology: A White Paper of the Italian Society of Medical and Interventional Radiology," *La Radiologia Medica* 128, no. 6 (2023): 755–764.
22. A. S. Tejani, H. Elhalawani, L. Moy, M. Kohli, and C. E. Kahn, "Artificial Intelligence and Radiology Education," *Radiology. Artificial Intelligence* 5, no. 1 (2023): e220084.
23. Y. Xu, Z. Jiang, D. S. W. Ting, et al., "Medical Education and Physician Training in the Era of Artificial Intelligence," *Singapore Medical Journal* 65, no. 3 (2024): 159.
24. J. Cervantes, B. Smith, T. Ramadoss, V. D'Amario, M. M. Shoja, and V. Rajput, "Decoding Medical Educators' Perceptions on Generative Artificial Intelligence in Medical Education," *Journal of Investigative Medicine* 72, no. 7 (2024): 633–639.
25. K. Pierre, M. Gupta, A. Raviprasad, et al., "Medical Imaging and Multimodal Artificial Intelligence Models for Streamlining and Enhancing Cancer Care: Opportunities and Challenges," *Expert Review of Anticancer Therapy* 23, no. 12 (2023): 1265–1279.
26. J. Mongan, J. Kalpathy-Cramer, A. Flanders, and L. M. George, "RSNA-MICCAI Panel Discussion: Machine Learning for Radiology From Challenges to Clinical Applications," *Radiology. Artificial Intelligence* 3, no. 5 (2021): e210118.
27. E. Buijs, E. Maggioni, F. Mazziotta, F. Lega, and G. Carrafiello, "Clinical Impact of AI in Radiology Department Management: A Systematic Review," *Radiologia Medica* 129, no. 11 (2024): 1656–1666, <https://doi.org/10.1007/s11547-024-01880-1>.
28. B. Koçak, A. Ponsiglione, A. Stanzione, et al., "Bias in Artificial Intelligence for Medical Imaging: Fundamentals, Detection, Avoidance,

- Mitigation, Challenges, Ethics, and Prospects,” *Diagnostic and Interventional Radiology* 31 (2024): 75–88.
29. M. G. Linguraru, S. Bakas, M. Aboian, et al., “Clinical, Cultural, Computational, and Regulatory Considerations to Deploy AI in Radiology: Perspectives of RSNA and MICCAI Experts,” *Radiology. Artificial Intelligence* 6, no. 4 (2024): 240225.
 30. P. Rouzrokh, J. E. Clarke, M. Hosseiny, M. Nikpanah, and M. Mokkarala, “Preparing Radiologists for an Artificial Intelligence – Enhanced Future: Tips for Trainees,” *Radiographics* 44, no. 8 (2024): 240042.
 31. A. L. Lindqwister, S. Hassanpour, P. J. Lewis, and J. M. Sin, “AI-RADS: An Artificial Intelligence Curriculum for Residents,” *Academic Radiology* 28, no. 12 (2021): 1810–1816.
 32. O. U. Hashmi, N. Chan, C. F. de Vries, A. Gangi, L. Jehanli, and G. Lip, “Artificial Intelligence in Radiology: Trainees Want More,” *Clinical Radiology* 78, no. 4 (2023): e336–e341.
 33. A. Hassankhani, M. Amoukhteh, P. Valizadeh, P. Jannatdoust, P. Sabeghi, and A. Gholamrezanezhad, “Radiology as a Specialty in the Era of Artificial Intelligence: A Systematic Review and Meta-Analysis on Medical Students, Radiology Trainees, and Radiologists,” *Academic Radiology* 31, no. 1 (2024): 306–321.
 34. N. V. Salastekar, C. Maxfield, T. N. Hanna, E. A. Krupinski, D. Heitkamp, and L. J. Grimm, “Artificial Intelligence/Machine Learning Education in Radiology: Multi-Institutional Survey of Radiology Residents in the United States,” *Academic Radiology* 30, no. 7 (2023): 1481–1487.
 35. M. Huisman, E. Ranschaert, W. Parker, et al., “An International Survey on AI in Radiology in 1041 Radiologists and Radiology Residents Part 2: Expectations, Hurdles to Implementation, and Education,” *European Radiology* 31, no. 11 (2021): 8797–8806.
 36. M. Cè, S. Ibba, M. Cellina, et al., “Radiologists’ Perceptions on AI Integration: An In-Depth Survey Study,” *European Journal of Radiology* 177 (2024): 111590.
 37. F. Schuur, M. H. Rezazade Mehrizi, and E. Ranschaert, “Training Opportunities of Artificial Intelligence (AI) in Radiology: A Systematic Review,” *European Radiology* 31, no. 8 (2021): 6021–6029.
 38. J. D. Perchik and S. Tridandapani, “AI/ML Education in Radiology: Accessibility Is Key,” *Academic Radiology* 30, no. 7 (2023): 1491–1492.
 39. G. Walsh, N. Stogiannos, R. van de Venter, et al., “Responsible AI Practice and AI Education Are Central to AI Implementation: A Rapid Review for All Medical Imaging Professionals in Europe,” *BJR Open* 5, no. 1 (2023): 20230033.
 40. A. S. Tejjani, R. M. Peshock, and K. M. Raj, “Evolving With Artificial Intelligence: Integrating Artificial Intelligence and Imaging Informatics in a General Residency Curriculum With an Advanced Track,” *Journal of the American College of Radiology* 21 (2024): 1608–1612.
 41. M. J. van Kooten, C. O. Tan, E. I. S. Hofmeijer, et al., “A Framework to Integrate Artificial Intelligence Training Into Radiology Residency Programs: Preparing the Future Radiologist,” *Insights into Imaging* 15, no. 1 (2024): 15.
 42. “European Training Curriculum,” 2024, <https://www.mysr.org/education/training-curricula/>.
 43. H. Arabi, A. AkhavanAllaf, A. Sanaat, I. Shiri, and H. Zaidi, “The Promise of Artificial Intelligence and Deep Learning in PET and SPECT Imaging,” *Physica Medica* 83 (2021): 122–137.
 44. D. Visvikis, C. Cheze Le Rest, V. Jaouen, and M. Hatt, “Artificial Intelligence, Machine (Deep) Learning and Radio(Geno)mics: Definitions and Nuclear Medicine Imaging Applications,” *European Journal of Nuclear Medicine and Molecular Imaging* 46, no. 13 (2019): 2630–2637.
 45. W. Shao, S. P. Rowe, and Y. Du, “Artificial Intelligence in Single Photon Emission Computed Tomography (SPECT) Imaging: A Narrative Review,” *Annals of Translational Medicine* 9, no. 9 (2021): 820.
 46. M. Fallahpoor, S. Chakraborty, B. Pradhan, et al., “Deep Learning Techniques in PET/CT Imaging: A Comprehensive Review From Sinogram to Image Space,” *Computer Methods and Programs in Biomedicine* 243 (2024): 107880.
 47. G. B. Bilgin, C. Bilgin, B. J. Burkett, et al., “Theranostics and Artificial Intelligence: New Frontiers in Personalized Medicine,” *Theranostics* 14, no. 6 (2024): 2367–2378.
 48. E. Yazdani, P. Geramifar, N. Karamzade-Ziarati, M. Sadeghi, P. Amini, and A. Rahmim, “Radiomics and Artificial Intelligence in Radiotheranostics: A Review of Applications for Radioligands Targeting Somatostatin Receptors and Prostate-Specific Membrane Antigens,” *Diagnostics* 14, no. 2 (2024): 181.
 49. T. Murata, T. Hashimoto, M. Onoguchi, et al., “Verification of Image Quality Improvement of Low-Count Bone Scintigraphy Using Deep Learning,” *Radiological Physics and Technology* 17, no. 1 (2024): 269–279.
 50. R. Hustinx, J. Pruim, M. Lassmann, and D. Visvikis, “An EANM Position Paper on the Application of Artificial Intelligence in Nuclear Medicine,” *European Journal of Nuclear Medicine and Molecular Imaging* 50, no. 1 (2022): 61–66.
 51. G. Currie, K. E. Hawk, and E. M. Rohren, “Ethical Principles for the Application of Artificial Intelligence (AI) in Nuclear Medicine,” *European Journal of Nuclear Medicine and Molecular Imaging* 47, no. 4 (2020): 748–752.
 52. G. Santo and A. Miceli, “Powering the Engine of Nuclear Medicine: Insights From the Next Generation Congress (Milan, 2024),” *European Journal of Nuclear Medicine and Molecular Imaging* 51 (2024): 3813–3814.
 53. S. A. Alowais, S. S. Alghamdi, N. Alsuhbeyany, et al., “Revolutionizing Healthcare: The Role of Artificial Intelligence in Clinical Practice,” *BMC Medical Education* 23, no. 1 (2023): 689.
 54. M. Kawamura, T. Kamomae, M. Yanagawa, et al., “Revolutionizing Radiation Therapy: The Role of AI in Clinical Practice,” *Journal of Radiation Research* 65, no. 1 (2024): 1–9.
 55. T. Ono, H. Iramina, H. Hirashima, T. Adachi, M. Nakamura, and T. Mizowaki, “Applications of Artificial Intelligence for Machine- and Patient-Specific Quality Assurance in Radiation Therapy: Current Status and Future Directions,” *Journal of Radiation Research* 65, no. 4 (2024): 421–432.
 56. E. Huynh, A. Hosny, C. Guthier, et al., “Artificial Intelligence in Radiation Oncology,” *Nature Reviews. Clinical Oncology* 17, no. 12 (2020): 771–781.
 57. J. Liao, X. Li, Y. Gan, et al., “Artificial Intelligence Assists Precision Medicine in Cancer Treatment,” *Frontiers in Oncology* 12 (2023): 998222.
 58. G. Francolini, I. Desideri, G. Stocchi, et al., “Artificial Intelligence in Radiotherapy: State of the Art and Future Directions,” *Medical Oncology* 37, no. 6 (2020): 50.
 59. S. Kalsi, H. French, S. Chhaya, et al., “The Evolving Role of Artificial Intelligence in Radiotherapy Treatment Planning—A Literature Review,” *Clinical Oncology* 36, no. 10 (2024): 596–605.
 60. N. D. Almeida, R. Shekher, A. Pepin, et al., “Artificial Intelligence Potential Impact on Resident Physician Education in Radiation Oncology,” *Advances in Radiation Oncology* 9, no. 7 (2024): 101505.
 61. Y. Song, J. Hu, Q. Wang, et al., “Young Oncologists Benefit More Than Experts From Deep Learning-Based Organs-at-Risk Contouring Modeling in Nasopharyngeal Carcinoma Radiotherapy: A Multi-Institution Clinical Study Exploring Working Experience and Institute Group Style Factor,” *Clinical and Translational Radiation Oncology* 41 (2023): 100635.

62. L. Boldrini, A. D'Aviero, F. De Felice, et al., "Artificial Intelligence Applied to Image-Guided Radiation Therapy (IGRT): A Systematic Review by the Young Group of the Italian Association of Radiotherapy and Clinical Oncology (yAIRO)," *Radiologia Medica* 129, no. 1 (2023): 133–151.
63. V. Bourbonne, A. Laille, N. Wagneur, Y. Ghannam, and A. Larnaudie, "Excitement and Concerns of Young Radiation Oncologists Over Automatic Segmentation: A French Perspective," *Cancers* 15, no. 7 (2023): 2040.
64. M. Guckenberger, N. Andratschke, M. Ahmadsei, et al., "Potential of ChatGPT in Facilitating Research in Radiation Oncology?," *Radiotherapy and Oncology* 188 (2023): 109894.
65. S. Siddique and J. C. L. Chow, "Artificial Intelligence in Radiotherapy," *Reports of Practical Oncology & Radiotherapy* 25, no. 4 (2020): 656–666.
66. C. Hurkmans, J. E. Bibault, K. K. Brock, et al., "A Joint ESTRO and AAPM Guideline for Development, Clinical Validation and Reporting of Artificial Intelligence Models in Radiation Therapy," *Radiotherapy and Oncology* 197 (2024): 110345.
67. L. Vandewinckele, M. Claessens, A. Dinkla, et al., "Overview of Artificial Intelligence-Based Applications in Radiotherapy: Recommendations for Implementation and Quality Assurance," *Radiotherapy and Oncology* 153 (2020): 55–66.
68. C. Mello-Thoms and C. A. B. Mello, "Clinical Applications of Artificial Intelligence in Radiology," *British Journal of Radiology* 96, no. 1150 (2023): 20221031.
69. "Radiology AI Register," 2024, radiology.healthairegister.com/.
70. B. S. Kelly, C. Judge, S. M. Bollard, et al., "Radiology Artificial Intelligence: A Systematic Review and Evaluation of Methods (RAISE)," *European Radiology* 32, no. 11 (2022): 7998–8007.
71. F. C. Kitamura, I. Pan, S. F. Ferracioli, K. W. Yeom, and N. Abdala, "Clinical Artificial Intelligence Applications in Radiology: Neuro," *Radiologic Clinics of North America* 59, no. 6 (2021): 1003–1012.
72. G. Chassagnon, C. De Margerie-Mellon, M. Vakalopoulou, et al., "Artificial Intelligence in Lung Cancer: Current Applications and Perspectives," *Japanese Journal of Radiology* 41 (2022): 235–244.
73. G. Chassagnon, M. Vakalopoulou, N. Paragios, and M. P. Revel, "Artificial Intelligence Applications for Thoracic Imaging," *European Journal of Radiology* 123 (2020): 108774.
74. B. M. Mervak, J. G. Fried, and A. P. Wasnik, "A Review of the Clinical Applications of Artificial Intelligence in Abdominal Imaging," *Diagnostics (Basel, Switzerland)* 13, no. 18 (2023): 2889.
75. J. H. Yoon, F. Strand, P. A. T. Baltzer, et al., "Standalone AI for Breast Cancer Detection at Screening Digital Mammography and Digital Breast Tomosynthesis: A Systematic Review and Meta-Analysis," *Radiology* 307, no. 5 (2023): e222639.
76. R. Y. L. Kuo, C. Harrison, T. A. Curran, et al., "Artificial Intelligence in Fracture Detection: A Systematic Review and Meta-Analysis," *Radiology* 304, no. 1 (2022): 50–62.
77. G. Keller, K. Rachunek, F. Springer, and M. Kraus, "Evaluation of a Newly Designed Deep Learning-Based Algorithm for Automated Assessment of Scapholunate Distance in Wrist Radiography as a Surrogate Parameter for Scapholunate Ligament Rupture and the Correlation With Arthroscopy," *La Radiologia Medica* 128, no. 12 (2023): 1535–1541.
78. V. Granata, R. Fusco, S. V. Setola, et al., "Machine Learning and Radiomics Analysis by Computed Tomography in Colorectal Liver Metastases Patients for RAS Mutational Status Prediction," *Radiologia Medica* 129, no. 7 (2024): 957–966, <https://doi.org/10.1007/s11547-024-01828-5>.
79. M. A. Mazurowski, M. Buda, A. Saha, and M. R. Bashir, "Deep Learning in Radiology: An Overview of the Concepts and a Survey of the State of the Art With Focus on MRI," *Journal of Magnetic Resonance Imaging* 49, no. 4 (2019): 939–954.
80. N. Daum, M. Blaivas, A. Goudie, et al., "Student Ultrasound Education, Current View and Controversies. Role of Artificial Intelligence, Virtual Reality and Telemedicine," *Ultrasound Journal* 16, no. 1 (2024): 44.
81. Y. T. Shen, L. Chen, W. W. Yue, and H. X. Xu, "Artificial Intelligence in Ultrasound," *European Journal of Radiology* 139 (2021): 109717.
82. E. P. V. Le, Y. Wang, Y. Huang, S. Hickman, and F. J. Gilbert, "Artificial Intelligence in Breast Imaging," *Clinical Radiology* 74, no. 5 (2019): 357–366.
83. T. Tsuboyama, M. Yanagawa, T. Fujioka, et al., "Recent Trends in AI Applications for Pelvic MRI: A Comprehensive Review," *La Radiologia Medica* 129, no. 9 (2024): 1275–1287, <https://doi.org/10.1007/s11547-024-01861-4>.
84. M. L. Richardson, E. R. Garwood, Y. Lee, et al., "Noninterpretive Uses of Artificial Intelligence in Radiology," *Academic Radiology* 28, no. 9 (2021): 1225–1235.
85. A. Hosny, C. Parmar, J. Quackenbush, L. H. Schwartz, and H. J. W. L. Aerts, "Artificial Intelligence in Radiology," *Nature Reviews. Cancer* 18, no. 8 (2018): 500–510.
86. D. T. Wagner, L. Tilmans, K. Peng, et al., "Artificial Intelligence in Neuroradiology: A Review of Current Topics and Competition Challenges," *Diagnostics* 13, no. 16 (2023): 2670.
87. J. Huo, X. Min, T. Luo, et al., "Computed Tomography-Based 3D Convolutional Neural Network Deep Learning Model for Predicting Micropapillary or Solid Growth Pattern of Invasive Lung Adenocarcinoma," *Radiologia Medica* 129, no. 5 (2024): 776.
88. G. Irmici, M. Cè, E. Caloro, et al., "Chest X-Ray in Emergency Radiology: What Artificial Intelligence Applications Are Available?," *Diagnostics* 13, no. 2 (2023): 216.
89. M. Yanagawa, R. Ito, T. Nozaki, et al., "New Trend in Artificial Intelligence-Based Assistive Technology for Thoracic Imaging," *La Radiologia Medica* 128, no. 10 (2023): 1236–1249.
90. S. Yan, J. Li, and W. Wu, "Artificial Intelligence in Breast Cancer: Application and Future Perspectives," *Journal of Cancer Research and Clinical Oncology* 149, no. 17 (2023): 16179–16190.
91. K. Lång, V. Josefsson, A. M. Larsson, et al., "Artificial Intelligence-Supported Screen Reading Versus Standard Double Reading in the Mammography Screening With Artificial Intelligence Trial (MASAI): A Clinical Safety Analysis of a Randomised, Controlled, Non-Inferiority, Single-Blinded, Screening Accuracy Study," *Lancet Oncology* 24, no. 8 (2023): 936–944.
92. G. Irmici, M. Cè, P. G. Della, et al., "Exploring the Potential of Artificial Intelligence in Breast Ultrasound," *Critical Reviews in Oncogenesis* 29, no. 2 (2024): 15–28.
93. Q. Ma, X. Lu, X. Qin, et al., "A Sonogram Radiomics Model for Differentiating Granulomatous Lobular Mastitis From Invasive Breast Cancer: A Multicenter Study," *La Radiologia Medica* 128, no. 10 (2023): 1206–1216.
94. M. Z. Yordanova, "The Applications of Artificial Intelligence in Radiology: Opportunities and Challenges," *European Journal of Medical and Health Sciences* 6, no. 2 (2024): 11–14.
95. G. Keller, A. Estler, J. Herrmann, et al., "Prospective Intraindividual Comparison of a Standard 2D TSE MRI Protocol for Ankle Imaging and a Deep Learning-Based 2D TSE MRI Protocol With a Scan Time Reduction of 48%," *La Radiologia Medica* 128, no. 3 (2023): 347–356.
96. D. De Santis, T. Polidori, G. Tremamunno, et al., "Deep Learning Image Reconstruction Algorithm: Impact on Image Quality in Coronary Computed Tomography Angiography," *La Radiologia Medica* 128, no. 4 (2023): 434–444.

97. C. P. Langlotz, "Will Artificial Intelligence Replace Radiologists?," *Radiology. Artificial Intelligence* 1, no. 3 (2019): e190058.
98. M. T. Contaldo, G. Pasceri, G. Vignati, L. Bracchi, S. Triggiani, and G. Carrafiello, "AI in Radiology: Navigating Medical Responsibility," *Diagnostics* 14, no. 14 (2024): 1506.
99. F. Jungmann, T. Jorg, F. Hahn, et al., "Attitudes Toward Artificial Intelligence Among Radiologists, IT Specialists, and Industry," *Academic Radiology* 28, no. 6 (2021): 834–840.
100. M. Braun, P. Hummel, S. Beck, and P. Dabrock, "Primer on an Ethics of AI-Based Decision Support Systems in the Clinic," *Journal of Medical Ethics* 47, no. 12 (2021): e3.
101. D. Visvikis, P. Lambin, K. Beuschaus Mauridsen, et al., "Application of Artificial Intelligence in Nuclear Medicine and Molecular Imaging: A Review of Current Status and Future Perspectives for Clinical Translation," *European Journal of Nuclear Medicine and Molecular Imaging* 49, no. 13 (2022): 4452–4463.
102. C. A. Mallio, A. C. Sertorio, C. Bernetti, and B. Beomonte Zobel, "Large Language Models for Structured Reporting in Radiology: Performance of GPT-4, ChatGPT-3.5, Perplexity and Bing," *Radiologia Medica* 128, no. 7 (2023): 808–812.
103. M. Nakajo, M. Jinguji, S. Ito, A. Tani, M. Hirahara, and T. Yoshiura, "Clinical Application of ^{18}F -Fluorodeoxyglucose Positron Emission Tomography/Computed Tomography Radiomics-Based Machine Learning Analyses in the Field of Oncology," *Japanese Journal of Radiology* 42, no. 1 (2024): 28–55.
104. A. Artesani, A. Bruno, F. Gelardi, and A. Chiti, "Empowering PET: Harnessing Deep Learning for Improved Clinical Insight," *European Radiology Experimental* 8, no. 1 (2024): 17.
105. S. A. Mirshahvalad, R. Eisazadeh, M. Shahbazi-Akbari, C. Pirich, and M. Beheshti, "Application of Artificial Intelligence in Oncologic Molecular PET-Imaging: A Narrative Review on Beyond ^{18}F -FDG Tracers – Part I. PSMA, Choline, and DOTA Radiotracers," *Seminars in Nuclear Medicine* 54, no. 1 (2024): 171–180.
106. R. Eisazadeh, M. Shahbazi-Akbari, S. A. Mirshahvalad, C. Pirich, and M. Beheshti, "Application of Artificial Intelligence in Oncologic Molecular PET-Imaging: A Narrative Review on Beyond ^{18}F -FDG Tracers Part II. ^{18}F -FLT, ^{18}F -FET, ^{11}C -MET and Other Less-Commonly Used Radiotracers," *Seminars in Nuclear Medicine* 54, no. 2 (2024): 293–301.
107. S. Bodapati, P. Abraham, A. Chen, et al., " ^{18}F -FES PET/CT Improves the Detection of Intraorbital Metastases in Estrogen-Receptor-Positive Breast Cancer: Two Representative Cases and Review of the Literature," *Tomography* 8, no. 2 (2022): 1060–1065.
108. L. Emmett, N. Papa, J. Buteau, et al., "The PRIMARY Score: Using Intraprostatic ^{68}Ga -PSMA PET/CT Patterns to Optimize Prostate Cancer Diagnosis," *Journal of Nuclear Medicine* 63, no. 11 (2022): 1644–1650.
109. R. Laudicella, A. Comelli, M. Schwyzer, et al., "PSMA-Positive Prostatic Volume Prediction With Deep Learning Based on T2-Weighted MRI," *Radiologia Medica* 129, no. 6 (2024): 901.
110. R. J. H. Miller and P. J. Slomka, "Artificial Intelligence in Nuclear Cardiology: An Update and Future Trends," *Seminars in Nuclear Medicine* 54, no. 5 (2024): 648–657.
111. R. J. H. Miller and P. J. Slomka, "Current Status and Future Directions in Artificial Intelligence for Nuclear Cardiology," *Expert Review of Cardiovascular Therapy* 22, no. 8 (2024): 367–378.
112. R. H. J. A. Slart, M. C. Williams, L. E. Juarez-Orozco, et al., "Position Paper of the EACVI and EANM on Artificial Intelligence Applications in Multimodality Cardiovascular Imaging Using SPECT/CT, PET/CT, and Cardiac CT," *European Journal of Nuclear Medicine and Molecular Imaging* 48, no. 5 (2021): 1399–1413.
113. D. Kusumoto, T. Akiyama, M. Hashimoto, et al., "A Deep Learning-Based Automated Diagnosis System for SPECT Myocardial Perfusion Imaging," *Scientific Reports* 14, no. 1 (2024): 13583.
114. R. Megna, M. Petretta, R. Assante, et al., "A Comparison Among Different Machine Learning Pretest Approaches to Predict Stress-Induced Ischemia at PET/CT Myocardial Perfusion Imaging," *Computational and Mathematical Methods in Medicine* 2021 (2021): 3551756.
115. A. J. Boyle, V. C. Gaudet, S. E. Black, N. Vasdev, P. Rosa-Neto, and K. A. Zukotynski, "Artificial Intelligence for Molecular Neuroimaging," *Annals of Translational Medicine* 9, no. 9 (2021): 822.
116. P. Alongi, R. Laudicella, F. Panasiti, et al., "Radiomics Analysis of Brain ^{18}F -FDG PET/CT to Predict Alzheimer's Disease in Patients With Amyloid PET Positivity: A Preliminary Report on the Application of SPM Cortical Segmentation, Pyradiomics and Machine-Learning Analysis," *Diagnostics* 12, no. 4 (2022): 933.
117. A. Bestetti, B. Zangheri, S. V. Gabanelli, V. Parini, and C. Fornara, "Union Is Strength: The Combination of Radiomics Features and 3D-Deep Learning in a Sole Model Increases Diagnostic Accuracy in Demented Patients: A Whole Brain ^{18}F -FDG PET-CT Analysis," *Nuclear Medicine Communications* 45, no. 7 (2024): 642–649.
118. V. Laskov, D. Rothbauer, and H. Malikova, "Robustness of Radiomic Features in ^{123}I -Ioflupane-Dopamine Transporter Single-Photon Emission Computer Tomography Scan," *PLoS One* 19, no. 4 (2024): e0301978.
119. G. Pagano, F. Niccolini, and M. Politis, "Imaging in Parkinson's Disease," *Clinical Medicine (London, England)* 16, no. 4 (2016): 371–375.
120. A. Chiaravalloti, R. Danieli, A. Lacanfora, B. Palumbo, C. Caltagirone, and O. Schillaci, "Usefulness of ^{18}F Flortetaben in Diagnosis of Alzheimer's Disease and Other Types of Dementia," *Current Alzheimer Research* 14, no. 2 (2017): 154–160.
121. G. Landry, C. Kurz, and A. Traverso, "The Role of Artificial Intelligence in Radiotherapy Clinical Practice," *BJR Open* 5, no. 1 (2023): 20230030.
122. G. Li, X. Wu, and X. Ma, "Artificial Intelligence in Radiotherapy," *Seminars in Cancer Biology* 86 (2022): 160–171.
123. Z. Hou, S. Gao, J. Liu, et al., "Clinical Evaluation of Deep Learning-Based Automatic Clinical Target Volume Segmentation: A Single-Institution Multi-Site Tumor Experience," *La Radiologia Medica* 128, no. 10 (2023): 1250–1261.
124. C. Xiao, J. Jin, J. Yi, et al., "RefineNet-Based 2D and 3D Automatic Segmentations for Clinical Target Volume and Organs at Risks for Patients With Cervical Cancer in Postoperative Radiotherapy," *Journal of Applied Clinical Medical Physics* 23, no. 7 (2022): e13631.
125. W. Liao, X. Luo, Y. He, et al., "Comprehensive Evaluation of a Deep Learning Model for Automatic Organs-at-Risk Segmentation on Heterogeneous Computed Tomography Images for Abdominal Radiation Therapy," *International Journal of Radiation Oncology, Biology, Physics* 117, no. 4 (2023): 994–1006.
126. A. Hosny, D. S. Bitterman, C. V. Guthrie, et al., "Clinical Validation of Deep Learning Algorithms for Radiotherapy Targeting of Non-Small-Cell Lung Cancer: An Observational Study," *Lancet Digital Health* 4, no. 9 (2022): e657–e666.
127. M. Zeverino, C. Piccolo, D. Wuethrich, et al., "Clinical Implementation of Deep Learning-Based Automated Left Breast Simultaneous Integrated Boost Radiotherapy Treatment Planning," *Physics and Imaging in Radiation Oncology* 28 (2023): 100492.
128. D. Lee, Y. c. Hu, L. Kuo, et al., "Deep Learning Driven Predictive Treatment Planning for Adaptive Radiotherapy of Lung Cancer," *Radiotherapy and Oncology* 169 (2022): 57–63.

129. L. Li, B. Xu, Z. Zhuang, et al., "Accurate Tumor Segmentation and Treatment Outcome Prediction With DeepTOP," *Radiotherapy and Oncology* 183 (2023): 109550.
130. "European Parliament P9_TA(2024)0138," Artificial Intelligence Act European Parliament Legislative Resolution of 13 March 2024 on the Proposal for a Regulation of the European Parliament and of the Council on Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts (COM(2021)0206-C9-0146/2021-2021/0106(COD)) (Ordinary legislative procedure: first reading), 2019.
131. M. Sallam, "ChatGPT Utility in Healthcare Education, Research, and Practice: Systematic Review on the Promising Perspectives and Valid Concerns," *Healthcare (Basel)* 11, no. 6 (2023): 887.
132. J. W. Gichoya, K. Thomas, L. A. Celi, et al., "AI Pitfalls and What Not to Do: Mitigating Bias in AI," *British Journal of Radiology* 96, no. 1150 (2023): 20230023.
133. H. J. W. L. Aerts, "The Potential of Radiomic-Based Phenotyping in Precision Medicine," *JAMA Oncology* 2, no. 12 (2016): 1636–1642.